Explainer: Browsers

Duke OLLI Digital Explorers SIG

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Learning Objectives

- What these things are: browser, site, page
- When actions you take with your browser result in data being generated/store locally vs on the site
- Have awareness of when a site is sending code to your browser to run locally
- Gain familiarity with common browser features

Agenda

1. Browsers, sites, & pages: what they are and how they work “under the covers”
   Also: key terms & review of URL (and links)
2. Browser UI overview
3. Common browser features
4. Fingerprints and footprints

What is a browser (a.k.a. web browser)?

- **David:** A type of app that you use to access sites on the Internet (or “web” or “World Wide Web”. Safari, Chrome, Edge, and Firefox are all browsers.
- wikipedia.org: A web browser, often abbreviated as browser, is an application for accessing websites. When a user requests a web page from a particular website, the browser retrieves its files from a web server and then displays the page on the user's screen. Browsers can also display content stored locally on the user's device.
- geeksforgeeks.org: The web browser is an application software used to explore the World Wide Web [a.k.a. the web]. It acts as a platform that allows users to access information from the Internet by serving as an interface between the client (user) and the server. The browser sends requests to servers for web documents and services, then renders the received HTML content, including text, images, links, styles, and scripts. Simply being connected to the Internet isn't enough; a browser is essential to search and view content online.
- computerhope.com: Alternatively called a web browser or Internet browser, a browser is a program to present and explore content on the World Wide Web. These pieces of content, including pictures, videos, and web pages, are connected using hyperlinks and classified with URIs (Uniform Resource Identifiers). This page is an example of a web page that can be viewed using a browser.

Most popular browsers

	Platforms:	Released in:	Owned by:	Global market
1. Google Chrome	Desktop + Mobile	2008	Alphabet, Inc. (USA)	~ 71.2%
2. Safari	Desktop + Mobile	2003	Apple Inc. (USA)	~ 14.8%
3. Microsoft Edge	Desktop + Mobile	2020	Microsoft Corporation (USA)	~ 4.6%
4. Firefox	Desktop + Mobile	2004	Mozilla Foundation (USA)	~ 2.3%
5. Samsung Internet	Mobile only	2012	Samsung Group (South Korea)	~ 1.9%
6. Opera	Desktop + Mobile	1995	Kunlun Tech Co., Ltd. (China)	~ 1.8%
7. Brave	Desktop + Mobile	2016	Brave Software, Inc. (USA)	~ 1.8%
8. DuckDuckGo	Desktop + Mobile	2019	DuckDuckGo, Inc. (USA)	~ 1.7%
9. Yandex Browser	Desktop + Mobile	2012	Yandex LLC (Russia)	~ 0.8%
10. UC Browser	Mobile only	2004	Alibaba Group (China)	~ 0.6%

Source: alltopeverything.com

Browser "Golden Rule"

- In general, all browsers load/render all pages and you can use any browser you prefer
- Occasionally, a page might not load/render correctly.
  Typical reasons this happens are:
  - Settings are not allowing page to use some browser or OS feature(s)
  - The page includes some code that your browser does not support

You can usually remediate these yourself

  - Site is "down"
  - Internet connection is "out"/"down"

You usually have to be patient until the site / internet service is restored

Definition of “site” (aka, “website” or “web site”)

- **David:** Think of a “site” as a place or location on the Internet that you can access (or “visit”) using a browser.
- britannica.com: collection of files and related resources accessible through the World Wide Web and the Internet via a domain name; organized around a "homepage" (or "landing page"), it is one of the foremost vehicles for mass communication and mass media
- godaddy.com: a collection of web pages that are grouped together and usually connected in various ways
- wix.com: a collection of web pages that allows individuals, businesses and organizations to share information, showcase services and connect with audiences online.
- hostinger.com: a place on the internet where you can keep information for others to see. This can be information about yourself, your business, or even topics of your interest. Based on the website category, people can also use them to shop, chat, study, and get entertained.
- computerhope.com: a central location of web pages that are related and accessed by visiting the website's home page using a browser.
- theknowledgeacademy.com: a digital platform accessible via the internet, serving as a virtual space where individuals, businesses, or organisations can create an online presence. It allows them to share information, offer various products or services, and connect globally. Websites come in multiple types and categories, each tailored to specific purposes and consisting of essential elements contributing to their functionality and design.
- intelivita.com: a bunch of web pages stored on a server that you can visit using a web browser

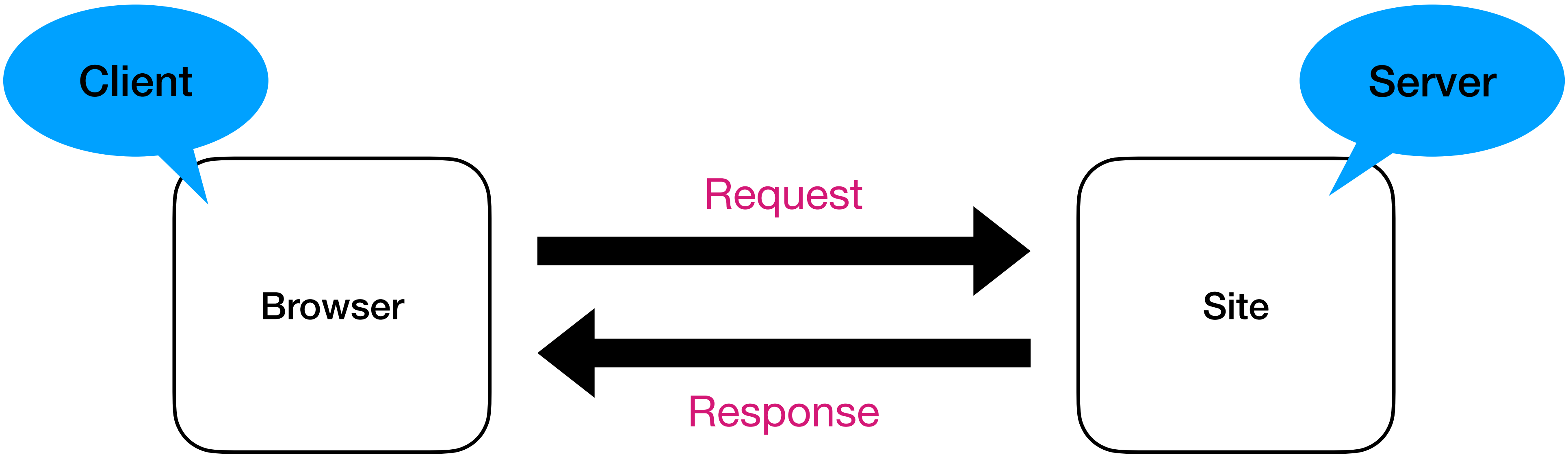
Definition of “page” (aka, “webpage” or “web page”)

- **David:** Think of a page as the content associated with a URL. i.e., a page is the content found at an Internet address.
- computerhope.com: A web page or webpage is a document, commonly written in HTML (hypertext markup language) viewed in an Internet browser. A web page can be accessed by entering a URL (uniform resource locator) address into a browser's address bar. A web page may contain text, graphics, and hyperlinks to other web pages and files.

  A web page provides information to viewers, including pictures or videos to help illustrate important topics. A web page may also be used as a method to sell products or services to viewers. Multiple web pages make up a website.
- techterms.com: Webpages are the files that make up the World Wide Web. An individual webpage is a text document written in HTML (hypertext markup language). When a web browser displays a webpage, it translates the markup language into readable content. Webpages are hosted on a web server, and the collection of webpages hosted on the same domain name make up a website.
- wikipedia.org: A web page, or webpage, is a Web document that is accessed in a web browser. A website typically consists of many web pages linked together under a common domain name. The term "web page" is therefore a metaphor of paper pages bound together into a book.
- merriam-webster.com: the block of information found at a single web address
- American Heritage Dictionary: A document on the World Wide Web, consisting of a hypertext file and any related files for scripts and graphics, and often hyperlinked to other documents on the Web.

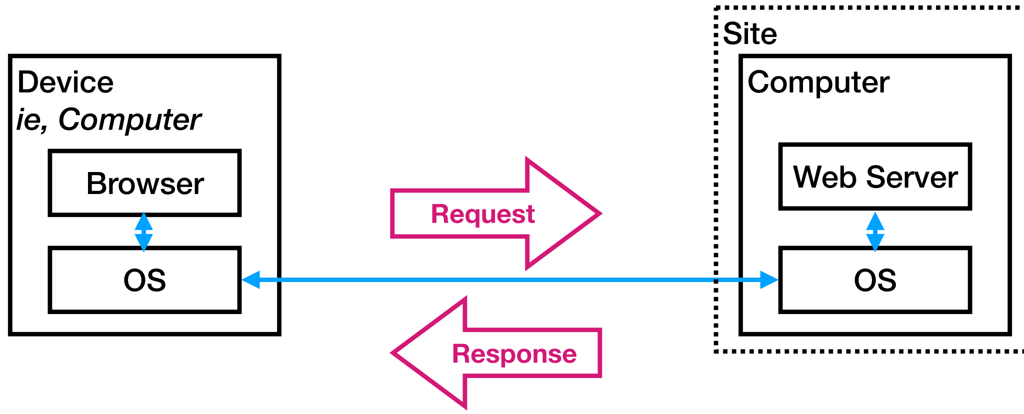
Types of Pages

Static	**Dynamic** (aka Interactive)
A page whose content does not change after it has been loaded by the browser. Example: all pages on our SIG site.	A page whose content changes when you interact with controls included on the page. Example: Amazon product pages
Responsive	**Non-Responsive**
A page that rearranges the layout of the content it includes based on the size of the screen or browser window. Example: all pages on our SIG site.	A page that “ignores” the screen/window size; you may have to scroll horizontally to view all the content on the page. Example: craiglist.org Raleigh/Durham page



In the Internet session of our Fall 2025 class, we used the above diagram to understand the basic interactions that occur between a browser and a site. Ignore (ie, forget about) the "client" and "server" bubbles; the concept I intended to describe with those bubbles during the Fall 2025 course turned out to be unhelpful/unnecessary.

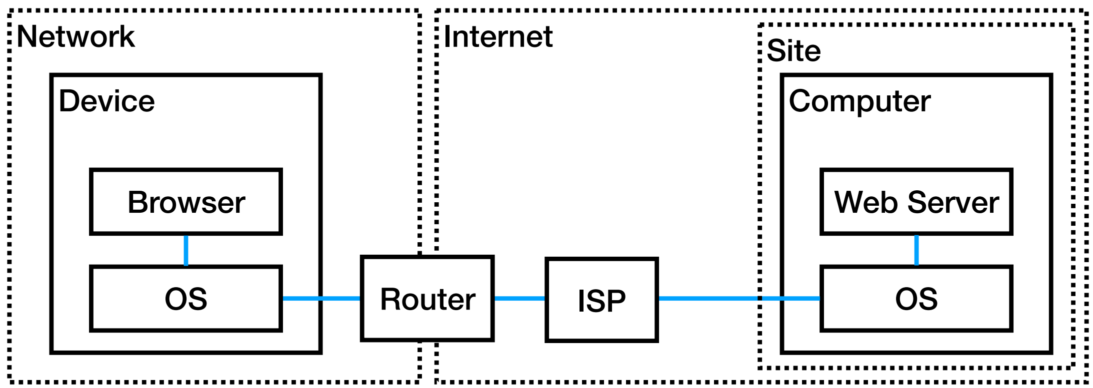
The following pages show diagrams that expand on/clarify the above diagram.



Key points

1. Your browser doesn’t “talk” to the Internet directly. Instead it “talks” directly to your operating system; the operating system directly handles interactions with the Internet on the browser’s behalf.

2. The sites you access through your browser are computer’s themselves.



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Key Terms

service **server** **host**	For our purposes, these terms are all synonyms for the word “site.” While there are differences between sites and services from the point of view of software engineers, from our point of view as users of the Internet and browsers, sites and services serve similar purposes in that they are the software “workhorses” residing on the Internet that satisfy the requests sent to them from our browsers.
home page	This term has two contexts. First, all sites have a home page; if you only enter a site’s name (e.g., facebook.com, youtube.com, wikipedia.org) in your browsers address bar, your browser will load that site’s home page into your browser window. Second, the content you see when you open a new window or tab in your browser is called your browser’s home page; there are browser options that let you set your browser’s home page.
URL	Essentially, the address of a page. A URL includes the name of a site and the name of a page at that site that you want to view. See the URL explainer on the DESIG site for more information.
link	A link (a.k.a. “hyperlink”) typically refers to an icon or text on a page that, when clicked or tapped, transfers the user to another part of the page or to another page on the current website or to a completely different website.
HTTP/HTTPS	An acronym for “Hypertext Transport Protocol”, HTTP is the software apps use to communicate with sites. HTTP is software itself; a copy of HTTP is included with operating systems. The “S” in HTTPS stands for “secure” and means that the communication between your app and a site is encrypted.
HTML	The programming language used to create pages. HTML determines the structure of pages. A page’s HTML determines how its content is layed out in your browser window.
cookie	A small “chunk” of data created by a web server while a user is browsing a website and stored on your device by the user's web browser. Cookies serve useful and sometimes essential functions on the web. They enable web servers to store stateful information (such as items added in the shopping cart in an online store) on the user's device or to track the user's browsing activity (including clicking particular buttons, logging in, or recording which pages were visited in the past).
IP address	A number that uniquely identifies your device when accessing the Internet. A devices’s IP address may be permanently assigned or supplied each time that it connects to the Internet by an Internet service provider. There are two different kinds of IP Addresses being used today: IPv4 and IPv6; IPv4 is being “phased out” in favor of IPv6 due to the exponential growth of the Internet.

Fingerprints & Footprints

- A page can “ask” your browser to send it a set of attributes about your environment (e.g., which browser you are using, which OS you are using, the size of your screen, etc); when taken together these attributes are unique relative to other Internet users. I.e., together the values of this set of attributes is like a **fingerprint**. Sites use these **fingerprints** to identify you when you visit a site and track your activity.
- When we access pages on sites, we tend to leave traces of our activity. E.g., for sites that allow/require you to sign in, you very specifically identify your self to the site and any activities you do on that site while signed in. Metaphorically, it is like leaving **footprints** in the sand when walking on the beach. The more sites you visit and accounts you create, the bigger your footprint. As your **footprint** grows larger, your exposure to having your privacy compromised grows; you security risks also increase as your **footprint** increases.

Common Browser Features

Search	You can easily search for things on the Internet using the Internet search engine that is coupled with your browser. While Google Search is the most popular Internet search engine (and coupled with most browsers by default) you can change the Internet search engine your browser uses.
Bookmark	If/when you find a page of interest to you that you to remember for use later, you can create a bookmark for that page to make it easy to access again later.
History	Your browser maintains a list of pages you visit/access and a list of Internet search queries you perform. Your browser uses these lists to simplify revisiting pages and Internet searches you previously performed.
Cache	Your browser saves some things to your local drive: cookies, browsing and search history, images, etc; browsers store these things to keep page load times short and maximize ease of use.
Autofill	When a page contains text boxes for you to fill in, your browser can fill in sometimes fill in these boxes for you. The kinds of text boxes in question usually have do with your personal information (name, address, phone number, etc.) How well this feature works with a given page depends in part on how the page was created.
Blockers	Browsers can attempt to inhibit certain types of advertising activities (namely, ad/marketing trackers and pop-up windows); when these features are enabled, your browser will likely block some but not all pop-up windows and ad/marketing trackers; also when these features are enabled, some pages may not load or work correctly.
Extensions	You can add capabilities to your browsers by installing extensions in your browser. These extensions include code that runs inside your browse, so you want to make sure any extensions you chose to install are trustworthy.
Privacy Mode	You can open browser windows in this “mode.” When your browser does not “remember” (i.e., save to your local disk) the pages you access, the internet searches you do, cookies, or cached files. This feature is called “Incognito Mode” with Chrome.
Tab Groups	You can create groups of pages to organize tabs you open those page in and then switch between groups.
Profiles	You can create separate environments (called “profiles”) for different types of activities/work you do with your browser. For example, you can create one profile for personal activities and a second profile for professional/academic work.
Reading Mode	Some browser settings/modes that remove visual distractions (like ads) on pages to help you focus on the “real” page content. This feature is sometimes also referred to as “distraction-free mode.”